

SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:
US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

Issuing Date 09-Feb-2024

Revision date 03-Dec-2025

Revision Number 1

1. Identification

Product identifier

Product Name Hammerhead Herbicide

Other means of identification

Product Code(s) 2122720

CAS Number MIXTURE

Synonyms None

Recommended use of the chemical and restrictions on use

Recommended Use Agriculture

Restrictions on use Do not use where contact with food or drinking water is possible Avoid prolonged contact with eyes, skin, and clothing

Details of the supplier of the safety data sheet

Supplier Address

ACETO LIFE SCIENCES, L.L.C.
4 Tri Harbor Court
Port Washington, NY 11050-4661
Phone: (516) 627-6000
Fax: (516) 627-6093
Email: aceto@aceto.com

Emergency telephone number

Emergency telephone number(s) ChemTrec: 1-800-424-9300
Outside of the United States: 1-703-527-3887

2. Hazard(s) identification

Classification

Acute toxicity - Oral	Category 4
Acute toxicity - Dermal	Category 4
Acute toxicity - Inhalation (Dusts/Mists)	Category 4
Skin corrosion/irritation	Category 2
Serious eye damage/eye irritation	Category 1
Skin sensitization	Category 1
Aspiration hazard	Category 1
Hazardous to the aquatic environment - acute	Category 2

Hazards not otherwise classified (HNOC)

Not applicable

Label elements



Danger

Hazard statements

Harmful if swallowed.
 Harmful in contact with skin.
 Harmful if inhaled.
 Causes skin irritation.
 Causes serious eye damage.
 May cause an allergic skin reaction.
 Toxic to aquatic life.
 May be fatal if swallowed and enters airways.

Precautionary Statements - Prevention

Wash face, hands and any exposed skin thoroughly after handling.
 Do not eat, drink or smoke when using this product.
 Wear protective gloves, protective clothing, eye protection and face protection.
 Avoid breathing dust, fume, gas, mist, vapors and spray.
 Use only outdoors or in a well-ventilated area.
 Contaminated work clothing must not be allowed out of the workplace.
 Avoid release to the environment.
 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
 Keep container tightly closed.
 Ground and bond container and receiving equipment.
 Use only non-sparking tools.
 Take action to prevent static discharges.
 Use explosion-proof electrical, ventilating and lighting equipment.

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
 Immediately call a POISON CENTER or doctor/physician.
 Call a POISON CENTER or doctor/physician if you feel unwell.
 If skin irritation or rash occurs: Get medical advice/attention.
 IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
 Wash contaminated clothing before reuse.
 IF INHALED: Remove person to fresh air and keep comfortable for breathing.

Call a POISON CENTER or doctor/physician if you feel unwell.
Rinse mouth.
IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician.
Do NOT induce vomiting.
In case of fire: Use CO₂, dry chemical, or foam for extinction.

Precautionary Statements - Storage

Store locked up.
Store in a well-ventilated place. Keep cool.

Precautionary Statements - Disposal

Dispose of contents and container in accordance with local, regional, national, and international regulations as applicable.

Other information

No information available.

3. Composition/information on ingredients**Substance**

Synonyms Not available

Formula Not available

Chemical name	CAS No.	Weight-%	Trade secret
Ethalfliuralin	55283-68-6	35.4	
Other Ingredients	N/A	64.6	

4. First-aid measures**Description of first aid measures**

Inhalation Remove to fresh air.

Eye contact Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.
Consult a physician.

Skin contact Wash skin with soap and water.

Ingestion Rinse mouth.

Most important symptoms and effects, both acute and delayed

Symptoms No information available.

Effects of Exposure No information available.

Indication of any immediate medical attention and special treatment needed

Note to physicians Treat symptomatically.

5. Fire-fighting measures

Suitable Extinguishing Media	Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	No information available.
Specific hazards arising from the chemical	No information available.
Hazardous Combustion Products	Carbon monoxide. Carbon dioxide (CO ₂). Hydrogen fluoride. Nitrogen oxides (NO _x). Hydrogen Chloride.
Explosion data	
Sensitivity to mechanical impact	None.
Sensitivity to static discharge	None.
Special protective equipment and precautions for fire-fighters	Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

Personal precautions	Ensure adequate ventilation.
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Methods and material for containment and cleaning up

Methods for containment	Prevent further leakage or spillage if safe to do so.
Methods for cleaning up	Pick up and transfer to properly labeled containers.

7. Handling and storage

Precautions for safe handling

Advice on safe handling	Handle in accordance with good industrial hygiene and safety practice.
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Conditions for safe storage, including any incompatibilities

Storage Conditions	Keep container tightly closed in a dry and well-ventilated place.
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8. Exposure controls/personal protection**Control Parameters****Exposure Limits**

This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

Chemical name	ACGIH TLV	OSHA PEL	NIOSH
Ethalfiuralin 55283-68-6	-	-	-
Other Ingredients N/A	-	-	-

Appropriate engineering controls**Engineering controls**

Showers
Eyewash stations
Ventilation systems.

Individual protection measures, such as personal protective equipment**Eye/face protection**

No special protective equipment required.

Skin and body protection

No special protective equipment required.

Respiratory protection

No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

General hygiene considerations

Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Information on basic physical and chemical properties

Physical state	Liquid
Appearance	Transparent Orange Gardner 13 liquid
Color	Transparent; Orange
Odor	Moderately Hydrocarbons
Odor Threshold	No information available

Property	Values	Remarks • Method
pH	5.06 (1% dispersion)	None known
pH (as aqueous solution)		None known
Melting point / freezing point	No data available	None known
Boiling point / boiling range	No data available	None known
Flash point	63 °C / 145 °F	None known
Evaporation rate	Not applicable	Not applicable
Flammability (solid, gas)	Not applicable	None known
Flammability Limit in Air		None known
Upper flammability or explosive limits	No data available	
Lower flammability or explosive limits	No data available	
Vapor Pressure	No data available	Not applicable
Relative vapor density	No data available	None known
Relative density	No data available	None known
Water solubility	No data available	Emulsifiable
Solubility in other solvents	No data available	None known
Partition coefficient: n-octanol/water		Not applicable
Autoignition temperature	No data available	No information available
Decomposition temperature		None known
Kinematic viscosity	Not applicable	None known
Dynamic viscosity	Capillary Viscosity: 4.50 centipoise @ 25C	None known

Other information

Explosive properties	No information available
Oxidizing properties	No information available
Softening point	No information available
Molecular weight	No information available
VOC content	No information available
Liquid Density	Density: 1.050 g/cm ³ (8.76 lb/gallon) @ 25C
Bulk density	No information available

10. Stability and reactivity

Reactivity	No information available.
Chemical stability	Stable under normal conditions.
Possibility of hazardous reactions	None under normal processing.
Conditions to avoid	None known based on information supplied.
Incompatible materials	None known based on information supplied.
Hazardous decomposition products	Nitrogen oxides (NOx). Carbon monoxide. Carbon dioxide (CO ₂). Hydrogen fluoride. Hydrogen Chloride.

11. Toxicological information

Information on likely routes of exposure

Inhalation	Avoid breathing dust, fume, gas, mist, vapors and spray.
Eye contact	Avoid contact with eyes.
Skin contact	Avoid contact with skin.
Ingestion	Do not take internally.

Symptoms related to the physical, chemical and toxicological characteristics

Symptoms No information available.

Acute toxicity

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Numerical measures of toxicity

No information available

The following ATE values have been calculated for the mixture

Oral LD50 > 4,000 mg/kg (rat)
Dermal LD50 > 5,000 mg/kg (rabbit)

Chemical name	Oral LD50	Dermal LD50	Inhalation LC50
Ethalfliuralin 55283-68-6	-	-	-
Other Ingredients N/A	-	-	-

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Skin corrosion/irritation No information available.

Serious eye damage/eye irritation No information available.

Respiratory or skin sensitization No information available.

Germ cell mutagenicity No information available.

Carcinogenicity No information available.

Chemical name	ACGIH	IARC	NTP	OSHA
Ethalfliuralin 55283-68-6	-	-	-	-
Other Ingredients N/A	-	cyclohexanone (108-94-1) 3	-	-

Reproductive toxicity No information available.

STOT - single exposure No information available.

STOT - repeated exposure	No information available.
Aspiration hazard	No information available.
Other Adverse Effects	No information available.
Interactive effects	No information available.

12. Ecological information

Ecotoxicity The environmental impact of this product has not been fully investigated.

Chemical name	Algae/aquatic plants	Fish	Toxicity to microorganisms	Crustacea
Ethalfliuralin 55283-68-6	-	-	-	-
Other Ingredients N/A	-	-	-	-

Persistence/Degradability No information available.

Bioaccumulative potential There is no data for this product.

Chemical name	Partition coefficient
Ethalfliuralin 55283-68-6	-
Other Ingredients N/A	-

Other adverse effects No information available.

13. Disposal considerations

Disposal methods

Waste from residues/unused products Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

Contaminated packaging Do not reuse empty containers.

14. Transport information

<u>DOT</u>	Not regulated
<u>TDG</u>	Not regulated
<u>MEX</u>	Not regulated
<u>ICAO (air)</u>	Not regulated
<u>IATA</u>	Not regulated
<u>IMDG</u>	Not regulated

15. Regulatory informationInternational Inventories

TSCA	Does not comply
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Chemical name	CAS No.	Inventory Listing Status	Commercial Activity Designation
Ethalfliuralin	55283-68-6	Present	Active
Other Ingredients	N/A	-	Unknown *

Chemical name	CAS No.	References
Ethalfliuralin	55283-68-6	-
Other Ingredients	N/A	-

DSL/NDL	Does not comply
EINECS/ELINCS	Does not comply
ENCS	Does not comply
IECSC	Does not comply
KECL	Does not comply
PICCS	Does not comply
AIIC	Does not comply
NZIoC	Does not comply

Legend:

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

EINECS/ELINCS - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

ENCS - Japan Existing and New Chemical Substances

IECSC - China Inventory of Existing Chemical Substances

KECL - Korean Existing Chemicals Inventory

PICCS - Philippines Inventory of Chemicals and Chemical Substances

AIIC - Australian Inventory of Industrial Chemicals

NZIoC - New Zealand Inventory of Chemicals

US Federal Regulations**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

CWA (Clean Water Act)

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

Chemical name	CWA - Reportable Quantities	CWA - Toxic Pollutants	CWA - Priority Pollutants	CWA - Hazardous Substances
Ethalfiuralin 55283-68-6	-	-	-	-
Other Ingredients N/A	-	-	-	-

CERCLA

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

Chemical name	Hazardous Substances RQs	Extremely Hazardous Substances RQs	Reportable Quantity (RQ)
Ethalfiuralin 55283-68-6	-	-	
Other Ingredients N/A	-	-	

US State Regulations**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

U.S. State Right-to-Know Regulations

This product does not contain any substances regulated under applicable state right-to-know regulations

Chemical name	New Jersey	Massachusetts	Pennsylvania
Ethalfiuralin 55283-68-6	-	-	-
Other Ingredients N/A	-	-	-

U.S. EPA Label Information

EPA Pesticide Registration Number Not applicable

EPA Statement

In the United States this product is regulated by the US Environmental Protection Agency (EPA) under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA). It is a violation of Federal law to use this product in a manner inconsistent with its labeling. read and follow all label directions. This product is excluded from listing requirements under EPA/TSCA. This chemical is a pesticide product registered by the Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required by Safety Data Sheets, and for workplace labels of non-pesticide chemicals.

16. Other information

NFPA **Health hazards** 1 **Flammability** 2 **Instability** 0 **Special hazards** -

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

TWA	TWA (time-weighted average)	STEL	STEL (Short Term Exposure Limit)
Ceiling	Maximum limit value	Sk*	Skin designation
+	Sensitizers		

Key literature references and sources for data used to compile the SDS

U.S. Agency for Toxic Substances and Disease Registry (ATSDR)
 U.S. Environmental Protection Agency ChemView Database
 European Food Safety Authority (EFSA)
 U.S. Environmental Protection Agency
 Acute Exposure Guideline Level(s) (AEGL(s))
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act
 U.S. Environmental Protection Agency High Production Volume Chemicals
 Food Research Journal
 Hazardous Substance Database
 International Uniform Chemical Information Database (IUCLID)
 Japan National Institute of Technology and Evaluation (NITE)
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)
 NIOSH (National Institute for Occupational Safety and Health)
 National Library of Medicine's ChemID Plus (NLM CIP)
 National Library of Medicine's PubMed database (NLM PUBMED)
 U.S. National Toxicology Program (NTP)
 New Zealand's Chemical Classification and Information Database (CCID)
 International Organization for Economic Co-operation and Development (OECD) Environment, Health, and Safety Publications
 International Organization for Economic Co-operation and Development (OECD) High Production Volume Chemicals Program
 International Organization for Economic Co-operation and Development (OECD) Screening Information Data Set
 United Nations World Health Organization (WHO)

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Revision Summary

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

End of Safety Data Sheet